

Tutorial 1

- Q1 (a) A schematic for an antenna azimuth position control system is shown in the Figure Q1a.

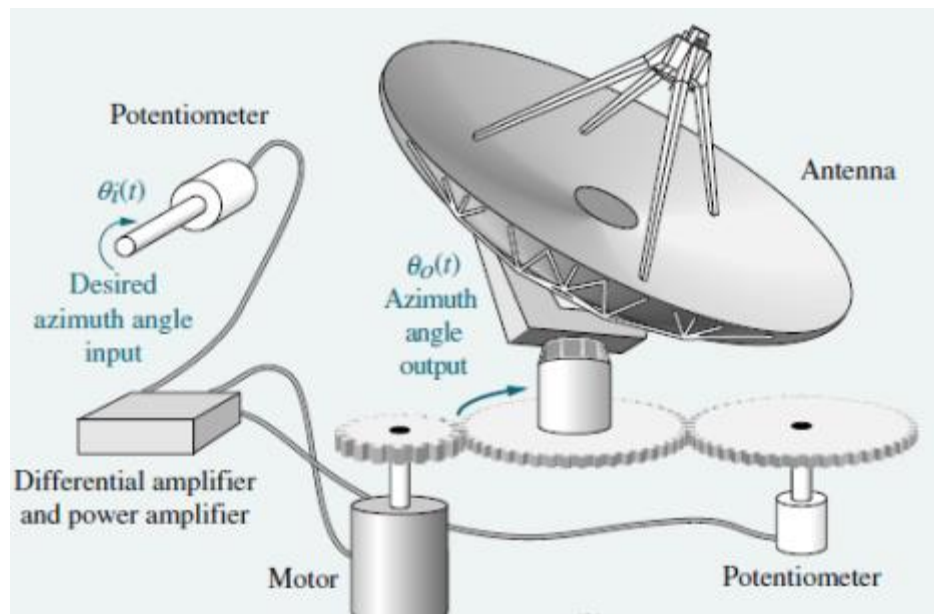


Figure Q1a

Draw a block diagram and explain the operation of the system. [7 marks]

- (b) Using block diagram reduction techniques, reduce the block diagram shown in figure Q1b to a single block. [18 marks]

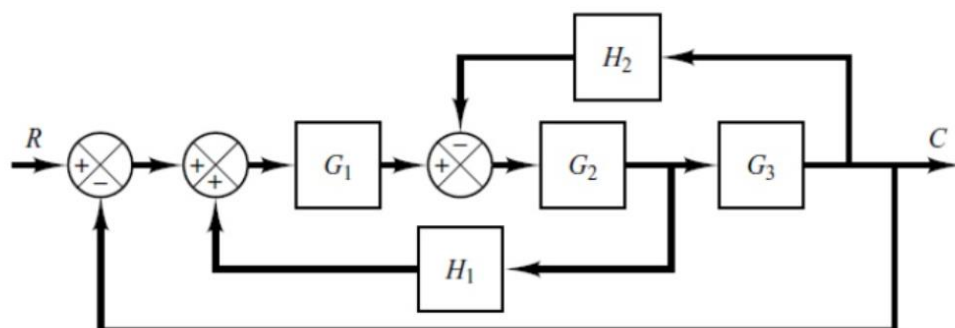


Figure Q1b

- Q2 (a) Apply block diagram reduction techniques to simplify the block diagram depicted in Figure Q2a into a single block.

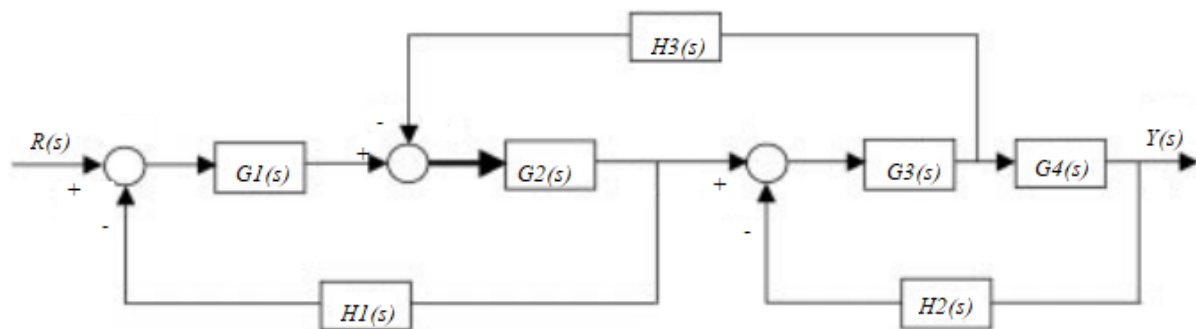


Figure Q2a

[18 marks]

- (b) How can block diagram reduction techniques be applied to optimise the performance of a control system? Provide insights into the key steps involved and discuss the potential impact on system stability and response. [7 marks]